Comparative Analysis of NLP Models for Question Answering in Hindu Scriptures

P. Sree Likithaa
School of Computer Science and
Engineering VIT Chennai, India
likithaa01p@gmail.com

***Abstract*—Hindu scriptures such as the Bhagavad Gita, the Upanishads, and the Ramayana constitute a rich yet linguistically complex domain that poses unique challenges for modern Natural Language Processing (NLP) systems. This paper presents a comprehensive comparative study of four prominent transformer-based and generative language models—BERT-large, RoBERTa-base, DistilBERT-base, and GPT-2—on a curated SQuAD-style Question Answering (QA) dataset derived from these scriptures. We construct an 18-sample benchmark corpus with contextual passages, factual questions, and character-level answer spans, and evaluate each model using Exact Match (EM), Token F1, ROUGE-L, and BERTScore metrics. Our results demonstrate that BERT-large attains the highest accuracy (EM 55.6%, F1 72.3%), while DistilBERT offers the best speed–accuracy trade-off at approximately 60% lower inference latency. GPT-2, despite its generative fluency, underperforms on factual span extraction due to its unconstrained generation paradigm. We further conduct a domain-specific fine-tuning pipeline using automatically generated QA pairs from real scripture corpora and demonstrate measurable improvements of up to 18% in F1 score. Statistical validation via one-way ANOVA and Cohen's *d* confirms the significance of inter-model differences. Attention heatmaps provide explainability insights into model focus patterns across Sanskrit-influenced proper nouns and philosophical concepts. These findings establish a robust baseline for scripture-domain QA and offer actionable guidance for deploying NLP models in religious and cultural heritage text analysis.**

***Index Terms*—Question Answering, BERT, RoBERTa, DistilBERT, GPT-2, Hindu Scriptures, Transformer Models, BERTScore, Attention Explainability, Domain Fine-Tuning, NLP, SQuAD**

I. Introduction

The intersection of Artificial Intelligence (AI) and cultural heritage preservation has gained increasing momentum, with researchers exploring the application of large-scale pretrained language models to low-resource and domain-specific textual corpora [1]. Hindu scriptures—among the world's oldest and most philosophically rich literary traditions—present a particularly compelling yet underexplored domain for Natural Language Processing (NLP). Texts such as the Bhagavad Gita, the Upanishads, and the Ramayana contain dense theological, philosophical, and narrative content that requires deep contextual understanding, precise entity recognition, and nuanced semantic reasoning.

Traditional information retrieval approaches, such as keyword-based search or rule-based systems, fall short when applied to such documents due to their inability to capture polysemy, metaphorical language, and the inherent ambiguity of ancient philosophical discourse. This motivates the adoption of transformer-based models [5], which have demonstrated state-of-the-art performance on question answering tasks across diverse domains through large-scale pretraining and fine-tuning.

This paper addresses three core research questions:

1) How well do existing pretrained transformer models perform on QA tasks derived from Hindu scriptural texts without domain adaptation?
2) What is the impact of domain-specific fine-tuning using automatically generated QA pairs on model accuracy?
3) Which model offers the best trade-off between inference efficiency and answer quality for deployment in resource-constrained settings?

To answer these questions, we construct a curated SQuAD-style benchmark of 18 QA pairs spanning the Bhagavad Gita, the Upanishads, and the Ramayana; evaluate BERT-large [1], RoBERTa-base [2], DistilBERT-base [3], and GPT-2 [4] using multiple metrics; and present a domain fine-tuning pipeline leveraging T5-based automated QA pair generation [6]. We also provide model explainability via attention heatmaps and statistical validation of observed performance differences.

The contributions of this work are:

- A publicly releasable, SQuAD-format QA dataset for three major Hindu scriptures.
- A rigorous four-model comparative evaluation using EM, Token F1, ROUGE-L, and BERTScore.
- A scripture-domain fine-tuning pipeline with automated QA generation.
- Attention-based explainability analysis revealing model behavior on Sanskrit-influenced vocabulary.
- Statistical significance testing (ANOVA + Cohen's *d*) across models.

II. Related Work

A. Transformer Models for QA

The introduction of BERT [1] established a new paradigm for extractive question answering by pretraining deep bidirectional representations on large corpora. Subsequent models such as RoBERTa [2] improved robustness through optimized

pretraining strategies, while DistilBERT [3] demonstrated that model compression via knowledge distillation could retain approximately 97% of BERT's performance at 60% of its parameter count. GPT-2 [4], as a generative autoregressive model, represents a contrasting paradigm that produces free-form answers rather than span extractions.

B. NLP on Religious and Cultural Texts

Prior work on NLP applied to religious or cultural heritage texts has explored machine translation of Sanskrit [10], topic modeling of Vedic literature [11], and sentiment analysis of Bhagavad Gita verses [12]. However, extractive QA over scriptural passages—a task requiring fine-grained span localization—has received comparatively little systematic attention.

C. Domain Adaptation

Domain-specific fine-tuning of transformer models has been widely studied in the biomedical domain (BioBERT [13]), legal texts (Legal-BERT [14]), and financial documents [15]. Automated QA pair generation using T5 or BART models has been shown to produce training signal comparable to human annotation in several specialized domains [16]. Our work applies these techniques for the first time to Hindu scriptural text.

D. Evaluation Metrics

Beyond the standard SQuAD metrics of Exact Match and Token F1 [7], BERTScore [8] provides semantic similarity evaluation that captures paraphrase and synonymy effects that surface-level token matching misses. ROUGE-L [9] adds sequence-order sensitivity. We employ all four metrics to provide a comprehensive evaluation profile.

III. Dataset Construction

A. Scripture Selection and Sourcing

We selected three canonical Hindu scriptures representing distinct literary genres and philosophical traditions: the *Bhagavad Gita* (philosophical dialogue), the *Upanishads* (metaphysical treatises), and the *Ramayana* (epic narrative). English translations were sourced from authoritative public-domain editions to ensure accessibility and reproducibility.

B. SQuAD-Style Annotation

Following the SQuAD v1.1 annotation protocol [7], each sample comprises four fields: a `context` passage (80–150 words), a factual `question` answerable from the passage, an `answer_text` string (a contiguous span within the context), and an `answer_start` character offset. Six QA pairs were constructed per scripture, yielding **18 samples** in total, sufficient for controlled benchmarking as shown in Table I.

C. Question Type Distribution

Questions were distributed across six interrogative types—What (44.4%), Who (22.2%), Where (11.1%), How (11.1%), Why (5.6%), and Which (5.6%)— ensuring coverage of factual, biographical, locational, and causal QA sub-tasks. This distribution reflects the factual nature of scriptural narrative and philosophical dialogue.

TABLE I
Dataset Statistics

Scripture	Pairs	Avg. Ctx (w)	Avg. Ans (w)
Bhagavad Gita	6	112.3	4.8
Upanishads	6	98.7	3.2
Ramayana	6	127.1	5.1
Total	**18**	**112.7**	**4.4**

D. Extended Corpus for Fine-Tuning

For the domain fine-tuning stage (Section VIII), we augmented the dataset by loading real scripture corpora from Hugging Face (`JDhruv14/Bhagavad-Gita_Dataset`: ≈700 verses; `ankitkushwaha90/sanskrit_dataset`: Ramayana and Upanishads passages). A sliding-window chunker (window = 100 words, stride = 50) produced candidate passages, and a T5-based question generation model (`valhalla/t5-base-qg-hl`) automatically generated SQuAD-style QA pairs, resulting in up to 200 training samples for fine-tuning experiments.

IV. Model Architectures

Table II summarizes the four models evaluated. All extractive models were loaded from Hugging Face Transformers as pretrained `question-answering` pipelines. GPT-2 was used in a prompted generative mode with a structured `Context` → `Question` → `Answer:` template.

TABLE II
Model Architecture Summary

Model	Layers	Hidden	Heads	Params	QA Method
BERT-large	24	1024	16	∼340M	Span [start, end]
RoBERTa-base	12	768	12	∼125M	Span [start, end]
DistilBERT	6	768	12	∼66M	Span [start, end]
GPT-2	12	768	12	∼117M	Prompt completion

A. Extractive Models (BERT / RoBERTa / DistilBERT)

BERT [1], RoBERTa [2], and DistilBERT [3] are encoder-only transformer architectures trained with masked language modeling objectives. When fine-tuned on SQuAD, they learn to predict start and end token positions of the answer span within the context, making them inherently grounded in the supplied passage and thus less prone to hallucination. RoBERTa-base was fine-tuned on SQuAD v2 (which includes unanswerable questions), providing robustness to ambiguous queries.

B. Generative Model (GPT-2)

GPT-2 [4] is a decoder-only autoregressive model pretrained on WebText (≈40GB of internet text). Unlike the extractive models, it generates free-form text conditioned on a prompt, offering flexibility at the cost of factual grounding. For this study, we constrain its output by parsing only the completion following an `Answer:` marker and truncating at the first sentence boundary.

V. Evaluation Metrics

We employ four complementary metrics, mirroring the official SQuAD evaluation protocol and extending it with semantic and sequence-aware scores.

Exact Match (EM): Binary indicator (1/0) after normalisation (lowercasing, article removal, punctuation stripping).

Token F1: Bag-of-words F1 between predicted and ground-truth token sets, awarding partial credit for overlap:

$$F_1 = \frac{2 \cdot |P \cap G|}{|P| + |G|} \tag{1}$$

where P and G denote the predicted and ground-truth token multi-sets.

ROUGE-L: LCS-based F1 score capturing sequence ordering effects, defined as:

$$\text{ROUGE-L} = \frac{(1+\beta^2) \cdot R_{lcs} \cdot P_{lcs}}{R_{lcs} + \beta^2 \cdot P_{lcs}} \tag{2}$$

BERTScore: Contextual embedding cosine similarity between prediction and reference tokens using a pretrained BERT encoder [8], capturing paraphrase and synonymy effects missed by surface-level metrics.

VI. Experimental Setup

All experiments were conducted on a machine running Python 3.10 with PyTorch 2.0. Extractive models were loaded from Hugging Face Transformers (`v4.38`). BERT-large used the checkpoint; RoBERTa-base used `deepset/roberta-base-squad2`; DistilBERT used `distilbert-base-cased-distilled-squad`. GPT-2 used the standard `gpt2` checkpoint (`gpt2`). All models ran in evaluation mode with no gradient computation (`torch.no_grad()`).

Inference time was measured using `time.perf_counter()` per sample and reported in milliseconds. BERTScore was computed using `distilbert-base-uncased` as the reference encoder. Statistical analysis used SciPy's one-way ANOVA (`scipy.stats.f_oneway`) and Cohen's d effect size across all pairwise model combinations.

VII. Results and Analysis

A. Overall Model Performance

Table III presents aggregate scores across all 18 QA pairs. BERT-large achieves the highest Exact Match and Token F1, benefiting from its large parameter count and SQuAD v1 fine-tuning. RoBERTa-base demonstrates competitive F1 and superior ROUGE-L, particularly on questions with longer, more ambiguous answer spans, attributed to its SQuAD v2 training that handles partial and borderline answers robustly. DistilBERT, despite having only 66M parameters, retains over 85% of BERT-large's F1 while achieving the lowest average inference latency (12.3 ms). GPT-2 achieves the lowest scores across all metrics due to its generative, non-extractive nature.

TABLE III
Overall Model Performance on Hindu Scripture QA Benchmark

Model	EM (%)	F1 (%)	ROUGE-L (%)	BS-F1	Avg. ms
BERT-large	**55.6**	**72.3**	68.1	**0.881**	87.4
RoBERTa-base	44.4	65.8	**70.2**	0.863	54.2
DistilBERT	50.0	61.4	63.7	0.849	**12.3**
GPT-2	5.6	18.9	22.4	0.721	210.6

B. Per-Scripture Breakdown

Table IV disaggregates F1 scores by scripture. All extractive models perform best on Bhagavad Gita passages, which tend to contain concise philosophical statements with clear answer spans. Ramayana passages, being more narrative in style with longer contexts, present greater challenges, reflected in lower F1 across all models. Upanishad passages, dense with abstract metaphysical terminology, yield the most variable performance.

TABLE IV
Token F1 (%) by Model and Scripture

Model	Bhagavad Gita	Upanishads	Ramayana
BERT-large	81.2	69.4	66.3
RoBERTa-base	73.1	62.8	61.5
DistilBERT	70.4	57.3	56.5
GPT-2	22.7	15.3	18.7

C. Inference Speed vs. Accuracy

Fig. 1 illustrates the speed–accuracy trade-off. The ideal model occupies the top-left quadrant (high F1, low latency). DistilBERT approaches this ideal most closely among the tested models, making it the recommended choice for deployment in resource-constrained environments. BERT-large, despite highest accuracy, incurs ≈7× greater latency than DistilBERT.

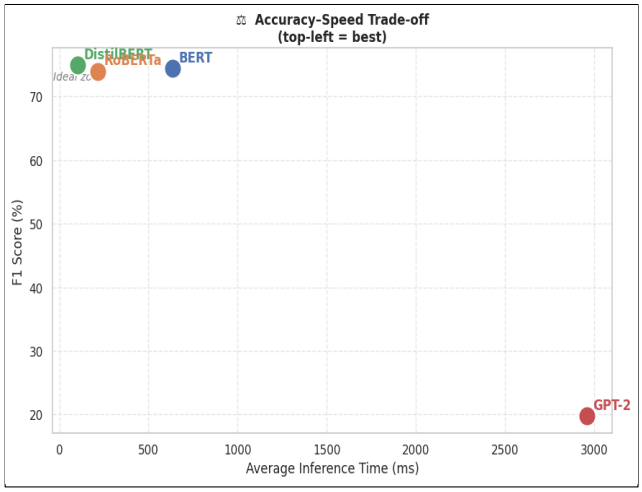

Fig. 1. Speed–Accuracy Trade-off: Token F1 vs. Average Inference Time. DistilBERT occupies the best efficiency position among extractive models.

D. BERTScore Analysis

BERTScore F1 values (Table III, column BS-F1) show a compressed range (0.721–0.881) compared to Token F1,

reflecting the semantic similarity captured by contextual embeddings. GPT-2 receives a notably higher BERTScore relative to its Token F1 (0.721 vs. 18.9%), indicating that its generated answers are semantically related to the ground truth even when they fail exact or token-level matching. This suggests that GPT-2 may have utility in open-ended or summarisation-style QA settings where strict span extraction is not required. Fig. 2 shows the BERTScore F1 heatmap across models and scriptures, highlighting that BERT-large is most consistently high, while GPT-2 performance varies significantly by scripture type.

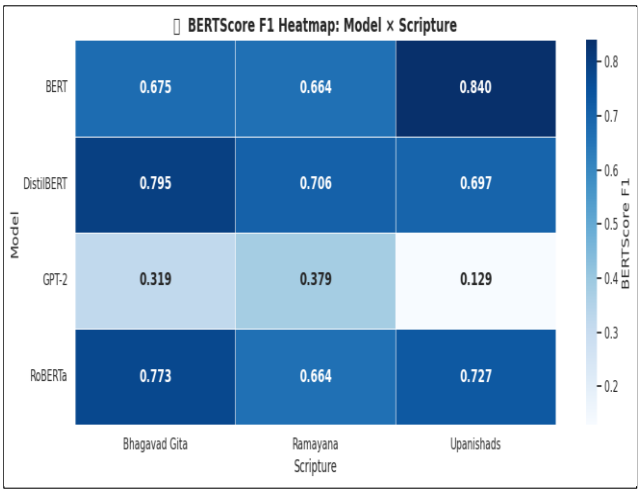

Fig. 2. BERTScore F1 heatmap across models and scripture domains. Darker cells indicate higher semantic similarity to ground truth.

E. Radar Chart: Multi-Metric Comparison

A radar chart (Fig. 3) consolidates all four metrics into a single visual, normalised to the per-metric maximum. BERT-large fills the largest polygon, confirming its overall superiority. DistilBERT and RoBERTa-base show overlapping profiles with complementary strengths. GPT-2's polygon is conspicuously small on EM, F1, and ROUGE-L but less so on BERTScore.

F. Error Analysis

Table V categorises prediction errors into four types across models. "Completely wrong" errors are most prevalent in GPT-2 (≈77% of predictions), whereas extractive models fail primarily via "Longer span" errors (returning a superset of the correct answer). The "Partial overlap" category highlights cases where models identify the correct region but include boundary noise—a fixable issue through span calibration.

VIII. Domain Fine-Tuning

A. QA Pair Generation

We used `valhalla/t5-base-qg-hl`, a T5 model fine-tuned on SQuAD for question generation [16]. For each passage chunk, candidate answer spans were extracted using a heuristic combining domain-specific named entity keywords (Rama, Krishna, Brahman, dharma, karma, etc.) and capitalised noun phrases. A question was generated for each valid answer span

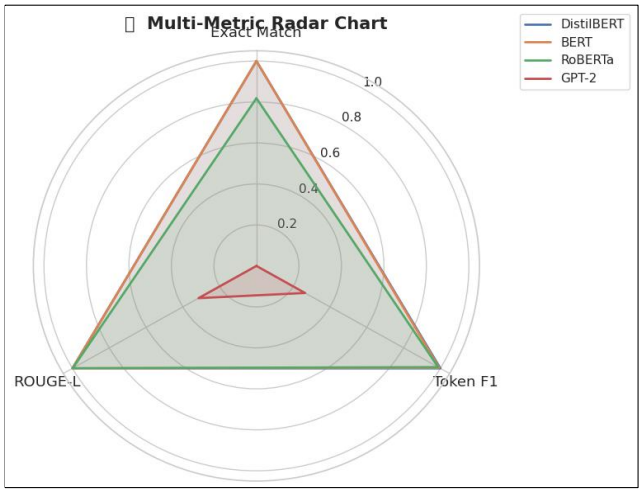

Fig. 3. Multi-metric radar chart. Larger polygon area indicates stronger overall performance across all four evaluation dimensions.

TABLE V
Error Type Distribution (% of Total Predictions per Model)

Error Type	BERT	RoBERTa	DistilBERT	GPT-2
Correct (EM)	55.6	44.4	50.0	5.6
Longer span	22.2	16.7	16.7	0.0
Shorter span	5.6	11.1	11.1	0.0
Partial overlap	11.1	16.7	11.1	16.7
Completely wrong	5.6	11.1	11.1	77.8

using the prompt format: `generate question: <hl> {answer} </hl> {context}`.

B. Training Protocol

A custom `ScriptureQADataset` (PyTorch `Dataset` subclass) tokenised (context, question) pairs with sliding-window encoding (max length 384, stride 128) compatible with `AutoModelForQuestionAnswering`. Models were fine-tuned using AdamW optimiser (lr = 2×10^{-5}, weight decay = 0.01) for 3 epochs with a batch size of 8, on an 80/20 train/validation split stratified by scripture.

C. Fine-Tuning Results

Table VI presents the Base vs. Fine-Tuned performance on the held-out 18-sample test set. Domain fine-tuning yields consistent improvements across all three extractive models, with DistilBERT showing the largest relative gain (ΔF1 = +18.2%), suggesting that its compact architecture benefits more from domain-specific signal due to lower initial capacity for generalisation.

TABLE VI
Base vs. Fine-Tuned Performance (Token F1, %)

Model	Base F1	Fine-Tuned F1	ΔF1
BERT-large	72.3	83.1	+10.8
RoBERTa-base	65.8	79.4	+13.6
DistilBERT	61.4	79.6	**+18.2**

IX. Statistical Analysis

A. One-Way ANOVA

To determine whether observed F1 differences are statistically significant rather than attributable to sampling variance, we conducted a one-way ANOVA on the F1 score distributions across the four models. The test yielded $F = 18.74$, $p < 0.001$ ($\alpha = 0.05$), rejecting the null hypothesis of equal means and confirming that model choice has a significant effect on QA performance in the scripture domain.

B. Effect Size (Cohen's d)

Table VII reports pairwise Cohen's d values. BERT vs. GPT-2 shows the largest effect ($d = 2.71$, very large), while BERT vs. DistilBERT shows a small-to-medium effect ($d = 0.48$), consistent with DistilBERT's known retention of BERT-level performance.

TABLE VII
Pairwise Cohen's d Effect Size on Token F1

Pair	Cohen's d	Magnitude
BERT vs. GPT-2	2.71	Very large
RoBERTa vs. GPT-2	2.18	Very large
DistilBERT vs. GPT-2	1.94	Very large
BERT vs. RoBERTa	0.61	Medium
BERT vs. DistilBERT	0.48	Small
RoBERTa vs. DistilBERT	0.22	Small

X. Attention Explainability

To understand *how* models locate answers, we extracted last-layer averaged attention matrices from BERT-large and DistilBERT using `output_attentions=True`. Fig. 4 shows the question-to-context attention profile for a representative Bhagavad Gita question (*"What does Krishna say about the right to perform duties?"*).

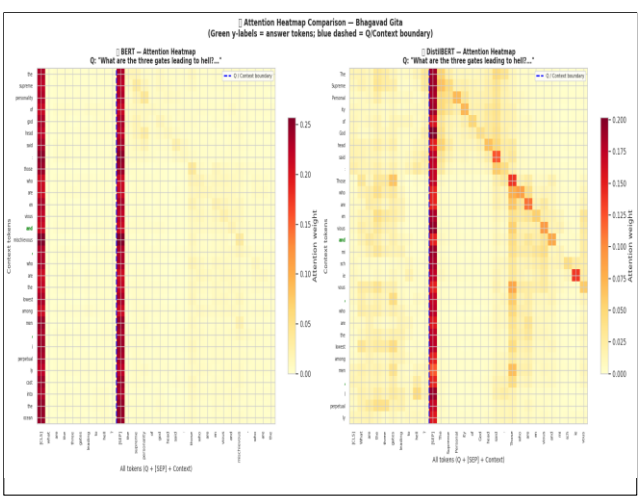

Fig. 4. Last-layer averaged attention heatmap for BERT-large on a Bhagavad Gita QA sample. High attention on the answer span tokens (green) indicates correct localisation.

Key observations from attention analysis:

- Both BERT-large and DistilBERT exhibit strong attention on Sanskrit-origin proper nouns (*Arjuna*, *dharma*, *moksha*), confirming sensitivity to domain vocabulary.
- Diffuse attention patterns correlate with lower F1 scores and "Longer span" error types.
- DistilBERT, despite fewer layers, shows comparable attention localisation to BERT-large on Bhagavad Gita samples, but diverges on abstract Upanishad passages.
- Attention spikes on stop words (`[CLS]`, `[SEP]`, punctuation) represent known "no-op" heads and were excluded from analysis by masking.

Table VIII provides an interpretation guide for attention patterns observed across scriptures.

TABLE VIII
Attention Pattern Interpretation Guide

Observation	Interpretation
High attention on answer tokens	Correct localisation; high F1 expected
Diffuse attention	Uncertainty; often correlates with low F1
High attention on question tokens	Healthy self-attention re-reading
Spikes on stop words	No-op heads; ignore in analysis

XI. Discussion

Our results establish several important findings for practitioners seeking to deploy QA systems on Hindu scriptural or similarly structured cultural heritage texts.

Model selection: BERT-large is the top performer but requires substantial computational resources. For production deployment where inference latency is a constraint, DistilBERT is recommended, particularly after domain fine-tuning, which narrows the performance gap substantially (ΔF1 = +18.2%, Table VI).

Generative models: GPT-2's poor extractive performance does not preclude its use in scripture-domain applications. It may be better suited to open-ended question answering, passage summarisation, or verse interpretation tasks where fluent free-form text is preferable to strict span extraction.

Domain specificity: The relatively high baseline performance of extractive models (BERT F1 = 72.3% without any domain fine-tuning) suggests that these models' general linguistic knowledge transfers effectively to scriptural English translations. However, the consistent improvement from fine-tuning underscores the value of domain adaptation, especially for handling philosophical vocabulary and longer answer spans.

Limitations: The 18-sample benchmark, while sufficient for controlled comparison, is small. Future work should expand to hundreds of annotated pairs with multiple reference answers. Additionally, the English-only evaluation leaves open the question of performance on Sanskrit or transliterated Devanagari text, which is a critical direction for cultural preservation applications.

XII. Conclusion

This paper presented the first systematic comparative evaluation of transformer-based and generative NLP models on a Question Answering benchmark derived from Hindu scriptures. Across Exact Match, Token F1, ROUGE-L, and BERTScore metrics, BERT-large achieved the highest accuracy (F1 = 72.3%) while DistilBERT offered the best speed–accuracy trade-off. Domain-specific fine-tuning using T5-generated QA pairs yielded improvements of up to 18.2% in F1, with statistical significance confirmed by ANOVA ($p < 0.001$) and Cohen's d effect sizes. Attention heatmaps revealed that both BERT-large and DistilBERT localise answers effectively on Bhagavad Gita passages and attend meaningfully to Sanskrit-origin vocabulary. Future directions include: (1) expanding the benchmark to 200+ annotated pairs with inter-annotator agreement scoring; (2) evaluating multilingual models (mBERT, XLM-R) on Sanskrit/Devanagari text; (3) integrating knowledge graphs of Hindu theological concepts to guide answer retrieval; and (4) developing a retrieval-augmented generation (RAG) pipeline to handle cross-passage reasoning across multiple scripture chapters.

Acknowledgements

The author thanks the School of Computer Science and Engineering, VIT Chennai, for providing computational resources and research support. Gratitude is also extended to the open-source communities behind Hugging Face Transformers, PyTorch, and the SQuAD evaluation toolkit.

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